

JÁN TÓTH

I am a systems programmer and game developer. Between 2021 and 2025, I was leading a small team of programmers at a startup by day, and developing a video game by night. At the moment, I am working on multiple smaller game prototypes.

I strive to maintain a sense of ownership in projects, as that translates into long-term productivity and good team morale. Ownership stems from familiarity with the codebase and the ability to modify it without fear. To that end, I responsibly evaluate the use of off-the-shelf tech and generative AI, so that the team does not unfavorably trade ownership for a short-term boost

When programming myself, I solve problems from first principles and try not to add artificial complexity. I do not shy away from diving into large and complex codebases (and I often do this for clients), but I try to make my own work simple to understand for others. I lean towards programming in C-like subsets of languages I work with: C++, Rust, JAI, C# and JavaScript.

Work Experience

Lead Software Engineer at Qubu

2021-2025 in Bratislava, Slovakia

Qubu is a software startup that aims to build quality housing for people by procedurally generating ready-to-build architecture. I was responsible for:

- Steering the company's software architecture across a 3D desktop application, CAD software plugins, build tools and asset pipelines.
- Mentoring a small team of talented programmers.
- Designing performant solutions from first principles. This included understanding how CPUs work, profiling, and reading assembly language as well as math papers.
- Implementing most of the desktop application's renderer.
- Building a SIMD ray-tracing light baker for evaluating insolation in buildings.
- Writing foundational components, such as threading primitives, allocators, pointer stable datastructures, internal developer tools and build tools.

Technology stack: Rust, C, C#, TypeScript, in-house engine, WebGPU, Rhino3D SDK.

Software Engineer at Subdigital

2017-2021 in Bratislava, Slovakia and remote

At Subdigital we created tools for computational designers, interactive demos, and interactive physical models. Of all our projects, I am most proud of the interactive model of the city Bratislava: <https://youtu.be/yGYEIrzWOGM?si=qAZZ-dkpcJYYt602>.

Technology stack: Rust, C, C#, TypeScript, multiple in-house engines, WebGPU, OpenGL, WebGL, Rhino3D SDK, Arduino.

Software Engineer at Vectary

2016-2017 in Bratislava, Slovakia

Helped build www.vectary.com, an accessible 3D modelling tool. My main contribution was simplifying the 3D editor's UI layer, reducing the size of the built bundle and simplifying the UI programming model.

Technology stack: TypeScript, JavaScript, React, Angular.js, Three.js, WebGL.

Software Engineer at V Lottery Systems

2015-2016 in Bratislava, Slovakia

Built high-performance server code for an e-gaming platform, mostly in the domain of scanning event logs looking for player cheating or other misbehavior.

Technology stack: Java, TypeScript, Spring Framework, Hikari CP, Angular.js.

Intern at Research Studios Austria

2014-2015 in Vienna, Austria

Prototyped natural language classification methods for the Web of Needs project, semantically matching related supply and demand items in a peer-to-peer online marketplace.

Technology stack: Java, Python, NumPy, SciPy, matplotlib.

Software Engineer at Davinci Software

2013-2014 in Bratislava, Slovakia

Researched and implemented named entity recognition algorithms on the Document Text Analyzer (DTA) product used to extract structured data out of PDF documents.

Technology stack: Java, JavaScript, Spring Framework, Esri OCR.

Software Developer at New Frontier Slovakia

2012-2013 in Bratislava, Slovakia

Helped develop a back-office solution for managing the VÚB Bank's product catalog.

Technology stack: Java, JavaScript, Spring Framework, Apache Tapestry.

Independent Consulting

embeddable.com

2026, remote

Analyzed and tested Cubestore to determine the root cause of scaling and stability issues the client was observing with the technology.

Technology stack: Rust.

Cruise (General Motors)

2021, remote

Optimized state management, rendering and CI pipelines of an internal browser-based app for building virtual reality test scenarios for self-driving cars. My main contributions were significantly reducing CI times (5x), initial application load times for large scenes (12x) and rendering performance during use, improving productivity of both users and the dev team.

Technology stack: TypeScript, React, WebGL.

Celestial Fruit on Earthly Ground

2021, remote

Built online virtual art installation exploring the colonial history of the banjo with Jonathan Reus.

Technology stack: TypeScript, Three.js, WebGL.

DUNGEONFOG GMBH

2020-2021 in Vienna, Austria and remote

Developed the early version of the renderer for Project Deios, the new iteration of the Dungeonfog map editor for pen and paper RPG enthusiasts.

Technology stack: Rust, C, JavaScript, Chromium Embedding Framework, WebGPU.

Listing Loop Pty Ltd

2018-2019, remote

With a small team built the early version of Listing Loop, a buyer focused real estate platform.

Technology stack: TypeScript, React, Firebase.

Volunteer Activity

Mentor at CoderDojo Bratislava

2022-now in Bratislava, Slovakia

I am a mentor in a children programming club. Instead of having a rigid curriculum, we give kids space to enjoy programming and learn computers driven purely by intrinsic motivation. Our mentees create all sorts of stuff: graphical and textual videogames, websites, simple hardware, and digital music and art.

Education

Computer Enhance

2023-now, computerenhance.com

Taking a course on how to make programs run fast on modern computers.

Msc in Cognitive Science

2013-2015, Comenius University in Bratislava

Research in natural language processing and computational neuroscience/psychology.

Mobility semester, project in natural language processing

2014, University of Vienna

Research in natural language processing and computational neuroscience/psychology.

Bsc in Applied Computer Science

2009-2012, Comenius University in Bratislava

Algorithms, data structures, computer science.

Ján Tóth

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Some of my code on Github (github.com/yanchith)

I occasionally stream programming on Twitch (www.twitch.tv/yanchibald)

Some of those streams get uploaded to YouTube (www.youtube.com/@yanchith)